

EDUCATION

- **Universitat Politècnica de Catalunya (UPC)** Barcelona, Spain
Ph.D. Candidate in Distributed Systems and Security Oct. 2025 – present
 - Transferred from UPF to finalize dissertation and defend.
- **Universitat Pompeu Fabra (UPF)** Barcelona, Spain
Ph.D. Researcher in Distributed Systems and Security under [Vanesa Daza](#) Oct. 2019 – Dec. 2024
 - Researched resource-constrained IoT ecosystems in blockchain environments, completing core preliminary studies and laying the foundational groundwork for the final dissertation.
- **Birla Institute of Technology and Science** Pilani, India
Masters of Engineering in Computer Science Aug. 2015– July. 2017
- **Maharshi Dayanand University** Rohtak, India
Bachelors of Technology in Computer Science Aug. 2009– July. 2013

EXPERIENCE

- **Advaita Labs** Singapore/Zurich, Remote
Research Lead at [Hetu](#) under Prof. [Jialin Li](#) Sept. 2024 – April 2026
 - **Core Research & PoCW Architecture:** Leading research on Distributed Knowledge Graphs and Intersubjective Consensus. Develop the [Proof of Causal Work \(PoCW\)](#) algorithm to establish attributable outputs in multi-agent environments, enabling reputation systems, AI alignment, and AI-native asset minting.
 - **Venture Validation & Academic Collaborations:** Drove applied research for the internal [venture arm](#), prototyping PoCs to validate emerging technologies and resolve complex engineering blockers. Fostered academic partnerships to translate theoretical models into practical, industry-ready applications.
- **Solayer Labs** San Fransico/Zurich, Remote
Protocols Researcher June 2024 - September 2024
 - **Developing Knowledge Repositories & Knowledge Assets:** At [Solayer Labs](#) developed research knowledge repositories and assets on scaling Solana's core tech. around Stake Weighted Quality of Service (SWQoS).
 - **Core Contributor:** Contributed code to Solayer's core repositories and assisted in developing funding-related documents, supporting both technical and strategic initiatives.
- **Status.im/Waku** Zug, Switzerland
Distributed Protocols Researcher September 2023 - February 2024
 - **Libp2p based Client:** [Developing Waku Store p2p client protocol](#) to extend features such as new message retention policies, extending the current database/store features to have a uniform view into Waku message types.
 - **RFC - DB Sync Store using Prolly Trees:** Developed and paper published a [state-of-the-art Prolly tree](#) for Sync protocol, enabling nodes to efficiently synchronize with their peer nodes atop libp2p, outperforming other implementations. An RFC was proposed to formalize and share this development with the broader community.
- **Google Summer of code(GSoC)** Remote
Student Collaborator June 2021 - Aug 2021
 - **HoneyPot:** [Developing an IoT/OT honeypot](#) to defend existing IoT/OT communication protocols such as coap, modbus, mqtt, etc. from cyber attacks. Modularly designed to supports other services such as ssh, http, echo etc.
 - **Multiple Interaction modes:** Supporting multiple interaction modes i.e. Local mode - protocols run locally, High mode - protocols run in separate containers and Hybrid mode - mix of both local and high mode.
- **Nokia Bell Labs** Paris, France
Research Engineering Intern, Blockchain Security Group Oct 2019 - Apr 2021
 - **Blockchain Security:** Blockchain Security project related to the IoT devices running on propriety blockchain solution.
 - **Lightweight Protocols:** Optimizing resource-intensive blockchain solutions for efficiency in both enterprise and public applications.

- Linkoping University** Linkoping, Sweden
 Research Assistant, under *Prof. Andrei Gurtov* *Feb 2019 - Aug 2019*
 - **Decentralized Systems:** Worked on decentralized communication between different electronic control units present in the vehicle.
 - **Profiling IoT:** Profiling of the IoT devices connected and reachable via the internet using tools such as Shodan and labelling the observation according to the known vulnerabilities dataset.
 - **IoT Risk Assessment:** Compiled report on IoT devices deployed in legacy systems and mission critical domains which could pose a greater risk if they are using vulnerable or unpatched protocols.
- EMC²(Now DelliEMC)** Pune, India
 Software Engineer, Protocols Group(SMB) *Jul 2017 - Jan 2019*
 - **SMB/CIFS Protocol:** Enhancing existing SMB/CIFS protocol on NAS box(M-tree based FS) which works for shared access of resources and serial ports between nodes on a network, widely used with Microsoft Windows.
 - **RFC Features:** Developing features for CIFS/SMB protocol referencing RFCs. Bypassing the open-source code and developing in-house protocols features. Continuous feature based delivery from SMB2 to SMB3 upgrade.
 - **Agile Practices:** As a Scrum master facilitated my team for developing, delivering, and sustaining complex products. Maintaining cross-functional collaboration with other teams and, the customer/end user.
- Google Summer of code(GSoC)** Remote
 Student Collaborator *May 2017 - Jul 2017*
 - **HPX:** Worked with HPX which is a C++ Standard Library for Concurrency and Parallelism. It implements all of the corresponding facilities as defined by the C++ Standard.
 - **Stack Overflow detection:** A stack overflow error can occur in a computer program using excessive memory. This could also be caused by segmentation fault. Simulated the detection with libsigsegv.
- EMC²(Now DelliEMC)** Pune, India
 Software Engineering Intern, DDOS Group *Jan 2017 - Jun 2017*
 - **DD OS Issues:** The Data Domain Operating System (DD OS) is the intelligence that powers Dell EMC Data Domain boxes used for data backup, archive and long term retention. Often faced issues in setup and network configuration both pre-work and in-working states.
 - **Services Debugger:** To avoid setup and services issues, developed a debugger that checks the client side setup and software configurations for smooth operations.
 - **Central Configurator:** Designed a python framework which monitors each and every service(bounded by a rule book) if they are setup and configured correctly. A security layer working in conjunction makes the product resist of manual errors and some network attacks. The product is now part of the DD OS across DelliEMC boxes.
- Navayuga Info. Spatial** New Delhi, India
 Application Development Intern *Summer 2013*
 - **E-commerce inventory:** E-commerce inventory management is the act of measuring the amount, location, pricing, and mix of products available from your business. Designed a mobile application that tracks the end to end business needs of items understocked, overstocked and out of stock
 - **Hybrid Application:** Developed a hybrid application in a native container that uses a mobile WebView object. When app starts, this object displays e-commerce items, using web technologies (JS, HTML5, CSS, MongoDB).

SELECTED PUBLICATIONS

1. Jayamine Alupotha, Mariarosaria Barbaraci, Ioannis Kaklamanis, Abhimanyu Rawat, Christian Cachin, and Fan Zhang, [Anonymous self-credentials and their application to single-sign-on](#), Cryptology ePrint Archive, Paper 2025/618 (2025).
2. Abhimanyu Rawat, Tarun Kumar Vangani, Hanno Cornelius, and Vanesa Daza, Accelerating proly trees: Simplified chunking for rapid updates, 6th Distributed Ledger Technology Workshop (DLT 2024) (2024).
3. Abhimanyu Rawat, Vanesa Daza, and Matteo Signorini, Not enough iot after all: Visiting transactional characteristics of iot blockchains, in *5th Distributed Ledger Technology Workshop (DLT 2023)* (2023).
4. Abhimanyu Rawat, Vanesa Daza, and Matteo Signorini, Offline scaling of iot devices in iot blockchain, *Sensors* **22**, [10.3390/s22041411](#) (2022).
5. Santiago Pagola Moledo, Abhimanyu Rawat, and Andrei Gurtov, **Best Paper Award** vendor-independent software-defined networking, in *2021 IEEE 2nd International Conference on Signal, Control and Communication (SCC)* (2021) pp. 168–174.

6. Abhimanyu Rawat, M. Khodari, M. Asplund, and Andrei Gurtov, Decentralized firmware attestation for in-vehicle networks, *ACM Trans. Cyber-Phys. Syst.* **5**, 10.1145/3418685 (2021).
7. D. Hasselquist, Abhimanyu Rawat, and A. Gurtov, Trends and detection avoidance of internet-connected industrial control systems, *IEEE Access* **7**, 155504–155512 (2019).
8. A. H. Sodhro, M. S. Obaidat, S. Pirbhulal, G. H. Sodhro, N. Zahid, and Abhimanyu Rawat, A novel energy optimization approach for artificial intelligence-enabled massive internet of things, in *2019 International Symposium on Performance Evaluation of Computer and Telecommunication Systems (SPECTS)* (2019) pp. 1–6.
9. M. Khodari, Abhimanyu Rawat, Mikael Asplund, and Andrei Gurtov, Decentralized firmware attestation for in-vehicle networks, in *Proceedings of the 5th on Cyber-Physical System Security Workshop*, CPSS '19 (Association for Computing Machinery, New York, NY, USA, 2019) p. 47–56.

AWARDS AND ACHIEVEMENTS

- **4 x Winner IC3 Blockchain Camp:** 2024, 2023, 2021 and 2020, **2 x Winner HackZurich** 2021 and 2022, **Winner IMDEA Hackathon** in Madrid 2020, **Winner BCN Hack** 1inch DeFi 2022
- **Academic achievement:** Recipient Doctoral Fellowship 60K USD/year from Protocol Las (Filecoin), Recipient Flashbots Research Grant, Recipient Sui Academic Research Awards, Stood in top 0.004% students among 10000 in BITSAT-HD exam, Merit Scholarship cum. 40% Fee Wavier at BITS Pilani, National Eligibility Test (NET-exam) qualified in first attempt, GATE qualified.
- **Others:** Ethereum Protocol Fellow May 2024, Reported payment discrepancy bug in [ccavenue](#), 3rd position in [Codility Titanium 2016 Challenge](#), Semifinalist in HackerRank University World Cup 2015, Certified Scrum Master, etc.

PROGRAMMING SKILLS

- **Languages: Proficient in:** C, C++, Python, Rust, Solidity, Go, bash, TeX **Basic to Moderate ability in:** Rust, Lisp, Java, JS, R
- **Industry Software Skills::** Wireshark, mininet, ns-2, Omnet++, Django, Flask, Android SDK, MySQL MongoDB, Docker, Raspberry Pi and more

EXTRA-CURRICULAR AND INTERESTS

- Fast Sudoku and Rubik's cube solving, flying glider planes, avid reader, into board-games, culinarian.

VISA REQUIREMENTS

- No Visa Sponsorship is required in Switzerland since I have a European Spouse.

REFERENCES

Available on Request